

Technical System Architecture & Operational Documentation

1. Executive Summary & Pipeline Overview

The **AI-Powered Document Screening Platform** is an enterprise-grade hybrid computer vision and cryptographic ledger system engineered for border control checkpoints. It processes multi-modal inputs—scanning physical travel documents alongside live passenger camera captures—to automate layout-agnostic segmentation, optical character recognition (OCR), multi-layered forgery detection, and biometric 1:1 face verification. The platform ensures zero-trust data integrity by coupling high-throughput neural inference with an immutable, permissioned consortium blockchain ledger.

2. Stage-by-Stage Module Specifications

Stage 0: Image Ingestion

The frontend interface ingests two discrete high-resolution streams simultaneously:

- **Document Image:** Physical travel document captured by a webcam/high-resolution camera.
- **Person's Photo:** Live camera capture of the traveler at the inspection booth.

Dual-Path Duplicate Routing

- **Original Image:** Forwarded directly to the Hash Engine.
 - **Copy of Original Image:** Routed to the pre-processing pipeline.
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Stage 1: Image Preprocessing

- **Background Segmentation:** BirefNet (a deep learning framework) is used to segregate the document substrate from background artifacts.
 - **Flattening & Perspective Warping:** Detects the contour of the document substrate (simplifying it to 4 or 6 polygonal edges) and applies an affine/perspective transformation matrix to unwarp and deskew the document canvas.
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Stage 2: YOLO Instance Segmentation Engine

The pre-processed image passes through a fine-tuned YOLO Segmentation model (`yolo11n-seg`) to extract **7 Universal Visual Classes**:

Class ID	Macro-Class Name	Target	Target Module
0	<code>doc_header</code>	Issuing state header & document title text	Hierarchical Router
1	<code>main_photo</code>	Primary facial portrait	Face Verification Engine
2	<code>ghost_photo</code>	Secondary laser-engraved / UV / shadow portrait	Photo-Substitution Engine
3	<code>mrz_zone</code>	Machine Readable Zone (2 or 3 lines of monospaced text)	Checksum MRZ
4	<code>signature</code>	Holder's handwritten or engraved signature	Stamp, Seal & Signature Module
5	<code>stamp_seal</code>	Official ink stamps, seals, or visa watermarks	Stamp, Seal & Signature Module
6	<code>fields_block</code>	Unified bounding block encompassing key-value personal fields	PaddleOCR Engine

Stage 3: Hierarchical Classification

- **Level 1 Classification (Document Type):** Categorizes the document layout into one of four macro-categories:
 - `passport`
 - `visa`
 - `national_id`
 - `dl` (Driving License)
- **Level 2 Classification (Country & Schema Resolution):** Parses country identifiers (e.g., characters 3–5 of MRZ Line 1 or header OCR) to execute *Identify country* and load *what fields to be extracted* from the local Schema Registry.

Stage 4: Verification and Checks Engine

The system executes three parallel analysis tracks on the extracted crops and pre-processed image canvas:

Track A: Hardcoded Logic Checks (Deterministic Engine)

- **MRZ Checksum Algorithm:** Executes ICAO Doc 9303 Modulus 10 calculation with a repeating 7-3-1 weighting sequence on:
 - Passport Number Check Digit
 - Date of Birth Check Digit
 - Date of Expiry Check Digit
 - Personal Number Check Digit
 - Composite Check Digit
- **Temporal & Date Logic Checks:**
 - Expiry Logic: Current Date < Date of Expiry
 - Issue Logic: Date of Birth < Date of Issue < Date of Expiry
 - Passport Lifespan: (Date of Expiry – Date of Issue) $\leq$ 10 years
- **String & Standard Format Checks:** Validates 3-letter issuing state codes against ISO 3166-1 alpha-3 standards and verifies character lengths (e.g., exactly 44 characters per line for TD3 passports).

Track B: ANN / DNN-Based Verifications

- **Face Verification Engine:**
 - Extracts deep facial embeddings using a facial recognition network (e.g., `InsightFace` / `DeepFace`).
 - Computes 1:1 cosine similarity across three vector pairs:

Embedding(main_photo) $\leftrightarrow$ Embedding(Person's Photo [Live])

Embedding(ghost_photo) $\leftrightarrow$ Embedding(Person's Photo [Live])

Embedding(main_photo) $\leftrightarrow$ Embedding(ghost_photo) (Catches photo-substitution attacks)

- **PaddleOCR Field-Value Extraction:**
 - Processes the pre-processed document for key-value pairs as per the schema resolved in Stage 3.
 - Applies spatial bounding box clustering to pair keys and values dynamically without cross-field contamination.
- **Stamp, Seal, and Signature Verification:**
 - Analyzes cropped `stamp_seal` regions via edge-density and color-histogram classifiers to distinguish physical ink stamps from printed digital graphics.
 - Evaluates `signature` regions for structural background editing artifacts.
- **MRZ and VIZ Cross-Matching:** Compares extracted Visual Inspection Zone (VIZ) text fields against decoded MRZ fields (e.g., confirming `VIZ_Surname == MRZ_Surname`).

Track C: Error Level Analysis (ELA)

- Re-compresses the image canvas and passes pixel difference heatmaps through a CNN to isolate compression anomalies, spliced text, and localized physical/digital alterations.
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Stage 5: Verification Pipeline & Dynamic Schemas

The extracted data and validation statuses are validated against the appropriate Dynamic Document Schema matching the document type.

Each schema dictates mandatory feature requirements:

- **passport_v1:** Standard non-biometric layout (Requires: `main_photo`, `mrz_zone`, `fields_block`).
 - **visa_v1:** Entry visa layout (Requires: `fields_block`, `stamp_seal`; Optional: `mrz_zone`).
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3. Cryptographic Ledger & Cybersecurity Infrastructure

1. Ingestion Hash Generation

Before passing the image to computer vision models, the system computes a SHA-256 cryptographic hash of the raw image bytes:

$$\text{Document_Hash} = \text{SHA256}(\text{Raw_Image_Bytes})$$

2. Pre-Execution Ledger Query

The system queries the shared ledger using `Document_Hash`:

- **Duplicate Detection:** If the hash exists and was previously marked **PASSED**, the system skips compute-heavy OCR/YOLO modules and routes directly to 1:1 Live Face Matching.
- **Revocation Check:** If the hash matches a record on the Stolen / Revoked Document Smart Contract, the transaction is immediately terminated and flagged as **CRITICAL HIGH RISK**.

3. Blockchain Transaction Logging

Once verification completes, the system commits an immutable transaction record to the ledger via smart contract execution.

4. Data Schemas, Scores, and Payload Specifications

A. Composite Risk Score Engine

The final output synthesizes scores into a normalized scale (0 to 100):

$$\text{Final_Risk_Score} = (w_1 \times S_{\text{mrz}}) + (w_2 \times S_{\text{face}}) + (w_3 \times S_{\text{forensics}}) + (w_4 \times S_{\text{logic}})$$

Score Category	Sub-Metrics Evaluated	Weight (w_i)	Pass Threshold
mrz_verification	Module 10 check digits, character length, B900 IR drop-out ratio	0.25	$\geq 95\%$
face_verification	Cosine similarity: (Main vs. Live), (Ghost vs. Live), (Main vs. Ghost)	0.35	$\geq 85\%$
other_scores (Forensics)	ELA discrepancy score, stamp/seal ink density, signature background continuity	0.20	$\geq 90\%$
other_scores (Logic)	Date validity (Expiry > Current), VIZ vs. MRZ text match, ISO country code	0.20	100% (Boolean)

Risk Action Matrix

- **Score 0 – 15 (LOW RISK):** Automated clearance (e-Gate open; total processing time < 3 seconds).
- **Score 16 – 50 (MEDIUM RISK):** Flagged for primary visual check by SSB officer (e.g., minor OCR read noise).
- **Score 51 – 100 (HIGH RISK):** Document rejected; automatic alert sent to secondary security booth.

B. JSON Payload Specifications

1. Internal Verification State Object (Python Engine Output)